

Unit 4, Lesson 3: Applying Exponent Laws

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

☐ I can apply the laws of exponents to generate equivalent numerical expressions.

4.3.1 Warm-Up: Negative Exponents

1. Place a digit in each blue box to make the statement true. Use only digits 0 to 9, at most one time each.

$$\square - \square = \frac{\square}{\square\square}$$

4.3.2 Collaboration: Applying More Than One Exponent Law

1. Fernanda and Junior were asked to write an equivalent expression of $\left(\frac{25^3}{25^8}\right)^{-1}$ by applying multiple exponent laws. Each student first described their plan.

- a. Complete each student's work based on the description of their plan.

Fernanda	Junior
I will first apply the quotient of powers exponent law. Then, I will apply the power of a power exponent law. Finally, I will re-write the decimal as a fraction and apply the negative exponent law.	I will first apply the power of a quotient exponent law. Then, I will apply the quotient of powers exponent law. Finally, I will re-write the decimal as a fraction and apply the negative exponent law.
$\left(\frac{25^3}{25^8}\right)^{-1}$ $= (25^3)^{-1}$ $= (25)^{-2}$ $= \left(\frac{5}{2}\right)^{-2}$ $=$	$\left(\frac{25^3}{25^8}\right)^{-1}$ $= \frac{(25)^{-3}}{(25)^{-8}}$ $=$ $=$ $=$

- b. Explain which method you prefer.

- c. Ask your partner which method they prefer. Summarize your partner's explanation.

4.3.3 Activity: Roundtable

In teams of three, assign a number (1–3) to each team member. Each student will complete the first step of their assigned problem and identify the exponent law used. Then, pass the problem to the student on your right to check the work and complete the next step in the problem.

For each new step, continue passing the problems to the student on your right until the problems have one base and one exponent.

Team Member 1: _____	
$\left(\left(\frac{4}{5}\right)^0 \cdot \left(\frac{5}{4}\right)^{-2}\right)^2$	<u>Exponent Law Used</u>
Team Member 2: _____	
$\frac{\frac{16}{25}}{\frac{16}{25}}$	<u>Exponent Law Used</u>

Team Member 3: _____	
$\frac{(-x)^3 \cdot 4}{(-x)^6 \cdot (-x)^{-2}}$	<u>Exponent Law Used</u>

4.3.4 Collaboration: Equivalent Expressions

1. Consider the expression $\frac{(\frac{1}{6})^3}{6^{-3}} \cdot 6^4$.

For each expression, work with your partner to determine if it is equivalent to $\frac{(\frac{1}{6})^3}{6^{-3}} \cdot 6^4$. Then, justify your choice by explaining using laws of exponents or showing your work.

Expression	Is it Equivalent to $\frac{(\frac{1}{6})^3}{6^{-3}} \cdot 6^4$?	Justify or Explain
a. $\frac{6^{-3}}{6^{-3}} \cdot 6^4$	☐ Yes ☐ No	
b. $6^{-8} \cdot 6^4$	☐ Yes ☐ No	
c. $(\frac{1}{6})^{-2} \cdot 6^4$	☐ Yes ☐ No	
d. $(\frac{1}{6})^{-8} \cdot 6^4$	☐ Yes ☐ No	
e. $\frac{(\frac{1}{6})^{-3}}{(\frac{1}{6})^3} \cdot (\frac{1}{6})^{-4}$	☐ Yes ☐ No	
f. 6^6	☐ Yes ☐ No	

4.3.5 Your Turn!

1. Apply the laws of exponents to write an equivalent numerical expression with one base and one positive exponent. Show your steps in the column on the left and identify the law of exponents used in each step in the column on the right.

a. $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$

Steps	Laws of Exponents

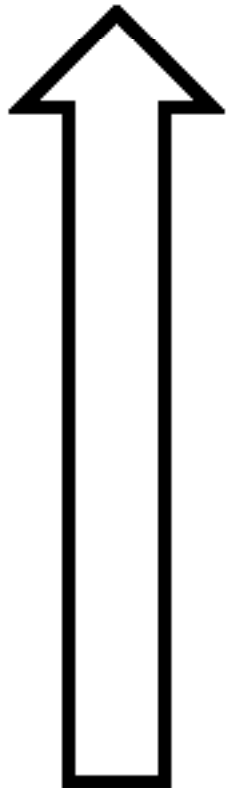
b. $\frac{5^4 \cdot 5^4}{5^9}$

Steps	Laws of Exponents

4.3.6 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of today's learning target.

- ☐ I can apply the Laws of Exponents to generate equivalent numerical expressions.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help with this next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.